Teacher:	Ray Joshua Balingkit	**Subject:**	Computing
Week:	6 (Term 1)	**Year:**	7
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will learn to demonstrate the taught binary and data-representation skills in a learning check. They will also learn to correct a binary or data-representation error and apply the correction to a new example. Through teacher demonstrations, discussion and practical activities, students will practise these skills and explain their ideas.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher observation, pupil explanations and the completed practical or paper activity. Record secure and developing outcomes in conversion, addition, overflow and representation; select a feedback target. Ask pupils to explain the correction and record who can now apply the skill without a prompt. Note pupils who need further guided practice.

LESSON SLOTS - MONDAY AND TUESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	28 September 2026	By the end of this lesson, students will be able to demonstrate the taught binary and data-representation skills in a learning check.	**H:** Use a brief retrieval prompt for bit, byte, place value and character code. **ID:** Explain the check: convert, add with carries and overflow, decode a character and describe pixels and sampling. **WD:** Model how to show working using an unrelated example without revealing the check's answers. **YD:** Pupils complete short written tasks and annotate an image or audio diagram, explaining one result. **C:** Record secure and developing outcomes in conversion, addition, overflow and representation; select a feedback target.	**Support:** Provide place-value grids, carry rows, a character-code table and labelled diagrams. Model one step at a time and allow spoken reasoning before writing. **Extension:** Ask pupils to justify a method, identify a deliberate error or test an eight-bit boundary, then verify the result using another representation.
2	29 September 2026	By the end of this lesson, students will be able to correct a binary or data-representation error and	**H:** Display an anonymous response with a misplaced carry or reversed place values.	**Support:** Provide place-value grids, carry rows, a character-code table and labelled diagrams. Model one step at a

		apply the correction to a new example.	**ID:** Reteach the common error using a grid, code table or diagram as appropriate. **WD:** Correct one example, explain each step and verify it by decoding or converting back. **YD:** Pupils correct their own responses and complete a similar independent task, with targeted guided practice where needed. **C:** Ask pupils to explain the correction and record who can now apply the skill without a prompt.	time and allow spoken reasoning before writing. **Extension:** Ask pupils to justify a method, identify a deliberate error or test an eight-bit boundary, then verify the result using another representation.
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Oxford International Lower Secondary Computing Student Book 7 printed pages 4-31
- Teacher computer and display
- Classroom computers or laptops and the class save folder
- Binary place-value grids, carry rows, character-code tables and pixel or sampling sheets
- Whiteboard and markers
- Student notebooks, pencils and task sheets
- Teacher reference: Unicode Consortium glossary, https://unicode.org/glossary/



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?